# MBA Fastrack 2025 (CAT + OMETs)

## QUANTITATIVE APTITUDE

**DPP: 1**

## Factors & Unit Digit Theorem

**Q1** Find the sum of all even factors of 2592

(A) 6116 (B) 7502
(C) 8116 (D) 8372

**Q2** Let $A = 2^{10} \times 3^6 \times 5^4$ and
$B = 2^7 \times 3^8 \times 7^5$.
Find the total number of natural numbers that are factors of at least one of $A$ or $B$.

(A) 817 (B) 761
(C) 580 (D) 538

**Q3** The $6^{th}$ position factor of 1800 from the beginning (when arranged in ascending order) is 6. If the $31^{st}$ position factor from the beginning is x. Find x

**Q4** Find the number of factors of 9,00,000 that are divisible by $10^3$ but not by $10^5$.

**Q5** Find the number of factors of $2^3 \times 3^2 \times 5^2 \times 7$ which are also the multiples of 56

**Q6** If the product of all divisors of $5^m$ equals $5^{190}$, determine the value of m.

**Q7** If N = $2^{42} \times 3^{34}$, How many factors of N, excluding 1, are perfect cubes?

**Q8** How many factors of 2,16,576 are divisible by 72?

(A) 12 (B) 14
(C) 24 (D) 28

**Q9** If 2PQ × RST = 226719, where each of P, Q, R, S, and T is a single-digit number, determine the value of (P+Q+R+S+T)

(A) 18 (B) 20
(C) 22 (D) 24

**Q10** If $m^2 + 100 = n^2$, how many integer pairs (m,n) satisfy this equation?

(A) 6 (B) 8
(C) 12 (D) 16

**Q11** The total number of factors of a number x and the total number of factors of 9x are in the ratio 4:5. Determine the highest power of 3 that is a factor of x.

(A) 2
(B) 5
(C) 7
(D) Cannot be determined

**Q12** Given:
p represents the number of integer pairs (m,n) that satisfy $m^2 - n^2 = 400$ for m,n N.
q represents the number of integer pairs (m,n) that satisfy $m^2 - n^2 = 400$ for m,n I.
Determine the value of $q^2 - p^4$.

**Q13** A = {2,2,2,2,3,3,4,4,9,9,25,25}
If two or more elements are selected from the given set and multiplied together, how many of these resulting products will be perfect squares?

**Q14** x, y and z are three natural numbers such that the product of x and y is 72 and that of y and z is 252. It is known that value of x is smaller than 12 then what is the largest possible sum of the three natural numbers?

(A) 60 (B) 39
(C) 45 (D) 76

**Q15** If 1AB×CDE = 124672 such that each of A, B, C, D and E are single digit numbers, what will be the value of ABCDE?

**Q16** How many even factors of ($4^3 \times 6^4 \times 15^2$) are greater than 48?

(A) 195 (B) 202

(C) 216 (D) 223

**Q17** How many factors of 1440 are not factors of 10800?

**Q18** If a and b are integers where ab=20, find the maximum possible value of

$\sqrt{43+(2-5a)(4-5b)}$.

(A) 23 (B) 28

(C) 31 (D) 37

**Q19** The number $897^{44}$ - $103^{44}$ is certainly divisible by:

(A) 49 (B) 81

(C) 112 (D) 125

**Q20** What will be the rightmost non-zero digit of $2370^{5320}$?

**Q21** If $(354x \times 7541 \times 4355 \times 8789 \times 5653)$ has 5 as unit digit, x can assume how many single digit values?

**Q22** Find the digit in the unit's place after solving the expression $38^{47} - 26^{39} - 17^{13}$

(A) 5 (B) 1

(C) 9 (D) 7

**Q23** What will be the unit digit for

$41^2 + 42^2 + 43^2 + \ldots + 100^2$?

(A) 0 (B) 1

(C) 3 (D) 5

**Q24** How many values can exist of the unit digit for $43^{(n-1)} \times 57^{(n+3)}$, if n is a natural number greater than 1?

(A) 1 (B) 2

(C) 3 (D) 4

**Q25** What will be digit at the unit's place for the given product?

$3^1 \times 3^2 \times 3^3 \times \ldots \times 3^{111}$

**Q26** Find the digit in the unit's place of ( $625^{33.75} \times 81^{12.25} \times 49^{11.5}$)?

**Q27** Find the unit digit of $27^{21!^{31!}}$.

**Q28** What will be the unit digit of $\frac{24^{42}}{6^{24}} + \frac{16^{60}}{64^{32}}$?

(A) 2 (B) 4

(C) 6 (D) 8

**Q29** If a = $\frac{27^{52}}{3^5} + \frac{16^{36}}{4^6}$, then find the last digit of a

**Q30** Find the units digit of expression :

$196^{143} + 174^{199} + 205^{218}$

(A) 0 (B) 2

(C) 5 (D) 7

# Answer Key

| | |
|---|---|
| Q1 | B |
| Q2 | B |
| Q3 | 300 |
| Q4 | 24 |
| Q5 | 9 |
| Q6 | 19 |
| Q7 | 179 |
| Q8 | B |
| Q9 | C |
| Q10 | A |
| Q11 | C |
| Q12 | 68 |
| Q13 | 58 |
| Q14 | C |
| Q15 | 28974 |
| Q16 | A |
| Q17 | 6 |
| Q18 | C |
| Q19 | D |
| Q20 | 1 |
| Q21 | 5 |
| Q22 | C |
| Q23 | A |
| Q24 | A |
| Q25 | 1 |
| Q26 | 5 |
| Q27 | 1 |
| Q28 | A |
| Q29 | 3 |
| Q30 | C |

# Hints & Solutions

Note: scan the QR code to watch video solution

**Q1 Text Solution:**

$$2592 = 2^5 \times 3^4$$

Sum of all even factors

$$\begin{aligned} &= \left(2^1 + 2^2 + 2^3 + 2^4 + 2^5\right) \\ &\times \left(3^0 + 3^1 + 3^2 + 3^3 + 3^4\right) \\ &= (2 + 4 + 8 + 16 + 32) \\ &\times (1 + 3 + 9 + 27 + 81) \\ &= (62 \times 121) \\ &= 7502 \end{aligned}$$

**Video Solution:**

**Q2 Text Solution:**

Number of factors = Prime number's (power +1)

Number of factors of A = (10+1) ×(6+1)×(4+1)

Number of factors of A $= 11 \times 7 \times 5 = 385$

Number of factors of B $= 8 \times 9 \times 6 = 432$

Factors common to A and B will be of the form of $= 2^{0\ to\ 7} \times 3^{0\ to\ 6}$

So number of factors common to A and B $= 8 \times 7 = 56$

So number of factors that divide atleast one of A or B $= 385 + 432 - 56 = 761$

**Video Solution:**

**Q3 Text Solution:**

1800 = $2^3 \times 3^2 \times 5^2$

Total number of factors of 1800 = 4 x 3 x 3 = 36

We know that the product of the $\mathbf{n}^{\text{th}}$ factor from the beginning and the $\mathbf{n}^{\text{th}}$ factor from the end is equal to the number itself.

Now $31^{\text{st}}$ factor from the beginning is nothing but $6^{\text{th}}$ factor from the end

Thus,

6x = 1800

x = 300

**Video Solution:**

**Q4 Text Solution:**

$900000 = 9 \times 100000 = 2^5 \times 3^2 \times 5^5$

Factors that will be divisible by $10^3$

$= \frac{(2^5 \times 3^2 \times 5^5)}{10^3}$

Factors that will be divisible by $10^3$ will be of the form $= 2^{3\ to\ 5} \times 3^{0\ to\ 2} \times 5^{3\ to\ 5}$

So number of such factors $= 3 \times 3 \times 3 = 27$

Factors that will be divisible by $10^5$

$= \frac{(2^5 \times 3^2 \times 5^5)}{10^5}$

Factors that will be divisible by $10^5$ will be of the form $= 2^{5\ \text{to}\ 5} \times 3^{0\ to\ 2} \times 5^{5\ \text{to}\ 5}$

So number of such factors $= 1 \times 3 \times 1 = 3$

So factors which are divisible by $10^3$ but are not divisible by $10^5 = 27 - 3 = 24$

**Video Solution:**

**Q5 Text Solution:**

$2^3 \times 3^2 \times 5^2 \times 7$

$= (8 \times 7) \times (3^2 \times 5^2)$

$= 56 \times (3^2 \times 5^2)$

Number of Factors of $(3^2 \times 5^2)$

$= (2+1)(2+1)$

$= 9$

**Video Solution:**

**Q6 Text Solution:**

$5^{190} = 5^0 \times 5^1 \times 5^2 \times \ldots \times 5^m$

$\Rightarrow 5^{190} = 5^{\frac{m(m+1)}{2}}$

$\Rightarrow \frac{m(m+1)}{2} = 190$

$\Rightarrow m\left(m+1\right) = 380$

$\Rightarrow m\left(m+1\right) = 19 \times 20$

$\Rightarrow m = 19$

**Video Solution:**

**Q7 Text Solution:**

Numbers of ways of choosing powers of 2 starting from 0 till all the multiples of 3 uptil 42 will be 15.

Similarly, the number of ways of choosing powers of 3 starting from 0 till all the multiples of 3 uptil 34 will be 12.

Thus, the number of perfect cube factors will be

15 x 12

= 180

Now, 1 needs to be excluded from the above.

Therefore, the required answer will be

180 - 1

= 179

**Video Solution:**

**Q8 Text Solution:**

$216576 = (216000 + 576)$
$= 9(24000 + 64)$
$= 2^6 \times 3^2 \times (375 + 1)$
$= 2^6 \times 3^2 \times 376$
$= 2^9 \times 3^2 \times 47$
Since
$72 = 2^3 \times 3^2$
factors which are divisible by 72 = $\frac{2^9 \times 3^2 \times 47}{2^3 \times 3^2}$
So, the factors which are divisible by 72 will be in the form of
$2^{3 \text{ to } 9} \times 3^{2 \text{ to } 2} \times 47^{0 \text{ to } 1}$
So, the number of such factors = 7*1*2 = 14
Ans

**Video Solution:**

**Q9 Text Solution:**

$2PQ \times RST = 226719$
$226719 = 3 \times 3 \times 3 \times 3 \times 3 \times 3 \times 311$
Since, $2PQ \neq 311$,
becuase first number is not same.
RST should be a multiple of 311.
Only power of 3 that is in the form of $2PQ$ is 243, so
$2PQ \times RST = 243 \times 3 \times 311 = 243 \times 933$
So
$P + Q + R + S + T = 4 + 3 + 9 + 3 + 3 = 22$

**Video Solution:**

**Q10 Text Solution:**

$m^2 + 100 = n^2$
$\Rightarrow n^2 - m^2 = 100$
$\Rightarrow (n + m)(n - m) = 100$
Factor pairs of 100 are
$(1, 100)$, $(2, 50)$, $(4, 25)$, $(5, 20)$ and $(10, 10)$
$(1, 100)$, $(4, 25)$ and $(5, 20)$ are pairs of odd and even numbers and will not result in integral values of m and n.
Since we are dealing with squares we need to consider negative values also, $(2, 50)$ will give us a total of 4 pairs
and $(10, 10)$ will give us a total of 2 pairs.
So total 6 cases are possible.

**Video Solution:**

**Q11 Text Solution:**

Let
$x = 2^a \times 3^b \times 5^c \times \ldots$
So
$9 \times x = 2^a \times 3^{b+2} \times 5^c \times \ldots$
So
$\frac{(a+1)(b+1)(c+1)\ldots}{(a+1)(b+3)(c+1)\ldots} = \frac{4}{5}$
$\Rightarrow \frac{(b+1)}{(b+3)} = \frac{4}{5}$
$\Rightarrow 5b + 5 = 4b + 12$
$\Rightarrow b = 7$
So
$x = 2^a \times 3^7 \times 5^c \times \ldots$
So the highest power of 3 that is factor of x is 7.

**Video Solution:**

**Q12 Text Solution:**

$m^2 - n^2 = 400$

$\Rightarrow (m+n)(m-n) = 400$

$400 = 1 \times 400 = 2 \times 200 = 4 \times 100$
$= 5 \times 80 = 8 \times 50 = 10 \times 40 = 16$
$\times 25 = 20 \times 20$

We cannot consider (odd, even) pairs as they will result in non integral values of m and n.

So

| (m-n) | (m+n) | m | n | p | q |
|---|---|---|---|---|---|
| ±2 | ±200 | ±101 | ±99 | 1 | 4 |
| ±4 | ±100 | ±52 | ±48 | 1 | 4 |
| ±8 | ±50 | ±29 | ±21 | 1 | 4 |
| ±10 | ±40 | ±25 | ±15 | 1 | 4 |
| ±20 | ±20 | ±20 | 0 (not natural) | 0 | 2 |

So total value of p = 4
and total value of q = 18
So

$q^2 - p^4 = 18^2 - 4^4 = 324 - 256 = 68$

**Video Solution:**

**Q13 Text Solution:**

The resulting numbers will be factors of

$2^4 \times 3^2 \times 4^2 \times 9^2 \times 25^2 = 2^8 \times 3^6 \times 5^4$

To identify only the perfect squares, lets express it as $(2^2)^4 \times (3^2)^3 \times (5^2)^2$ ( so as to obtain products as perfect squares)

So number of resultant perfect square products = (4+1)×(3+1)×(2+1) = 60

But we need to remove 1 and 25 as resultant product as no two elements of the set A will give us a product of 1 or 25.

So possible number of perfect square products = 60-2 = 58

**Video Solution:**

**Q14 Text Solution:**

$x \times y = 72$

$y \times z = 252$

So

$\frac{x \times y}{y \times z} = \frac{72}{252}$

$\Rightarrow z = 3.5x$

We have to find the maximum value of x+y+z= x+y+3.5x = 4.5x+y

Since x and y give a product of 72, they must be factors of 72

Factors of 72 = 1,2,3,4,6,8,9,12,18,24,36,72

Since x is smaller than 12, possible values of x are 1,2,3,4,6,8 and 9 but z=3.5x, since z is also a natural number, x should be an even number or else it will result in z being a decimal.

So the only possible values of x in this case are 2,4,6 and 8

| x | y | 4.5x+y |
|---|---|---|
| 2 | 36 | 45 |
| 4 | 18 | 32 |
| 6 | 12 | 39 |
| 8 | 9 | 45 |

So the largest possible sum of x, y and z is 45.

**Video Solution:**

**Q15 Text Solution:**

Given

$1AB \times CDE = 124672$

$Factori\sin g$

$124672 = 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 487$

$Since\ 1AB\ is\ smaller\ than\ 487,\ 487\ will\ be\ a\ factor\ of\ CDE,\ and\ 1AB\ should\ be\ a\ power\ of\ 2$

$So,\ only\ power\ of\ 2\ with\ 1\ at\ hundreds\ place\ is\ 2^7 = 128$

$So$

$1AB \times CDE = 128 \times 974 = 124672$

So, ABCDE=28974

**Video Solution:**

**Q16 Text Solution:**

$4^3 \times 6^4 \times 15^2 = 2^{10} \times 3^6 \times 5^2$

Total even factors of $4^3 \times 6^4 \times 15^2$ =

$10 \times 7 \times 3 = 210$

$48 = 2^4 \times 3$

Powers of 2 less than 48 = 2, 4, 8, 16, 32

Product of powers of 3 with powers of 2, less than or equal to 48 = 6, 12, 24, 48, 18, 36

Product of powers of 5 with powers of 2, less than or equal to 48 = 10, 20, 40

Product of powers of 3 with powers of 5 with powers of 2, less than or equal to 48 = 30

So total even factors less than or equal to 48 = 15

So even factors greater than 48 = 210 - 15 = 195

**Video Solution:**

**Q17 Text Solution:**

We know that,

$1440 = 2^5 \times 3^2 \times 5$

$10800 = 2^4 \times 3^3 \times 5^2$

We can find the common factors of 1440 and 10800. For that, we have to find the highest common powers of all the common prime factors in 1440 and 10800.

The common factors from 1440 and 10800 =

$2^4 \times 3^2 \times 5$

So, total number of common factors from this = (4 + 1)(2 + 1)(1 + 1) = 30

Also, the number of factors of 1440 = (5 + 1)(2 + 1)(1 + 1) = 36

Hence, the number of factors of 1440 which are not factors of 10800 = 36 - 30 = 6

**Video Solution:**

**Q18 Text Solution:**

$43+\left(2-5a\right)\left(4-5b\right)=43+8-20a$
$-10b+25ab=551-10\left(2a+b\right)$

$\sqrt{43+(2-5a)(4-5b)}$ will be maximum when $43+\left(2-5a\right)\left(4-5b\right)$ will be maximum whiich means when $551-10\left(2a+b\right)$ will be maximum.

To maximize it we need to minimize $\left(2a+b\right)$.

$20=2^2\times 5$

So total $3\times 2=6$ pairs are possible considering positive factors and in total 12 pairs are possible including negative pairs.

However since we need to minimize the sum here we will consider only the negative factors.

| a×b | 2a+b |
|---|---|
| -1×-20 | -22 |
| -2×-10 | -14 |
| -4×-5 | -13 |
| -5×-4 | -14 |
| -10×-2 | -22 |
| -20×-1 | -41 |

So minimum value of

$\sqrt{43+(2-5a)(4-5b)}$ is

$=\sqrt{551-10(-41)}$

$=\sqrt{961}$

$=31$

**Video Solution:**

**Q19 Text Solution:**

We know that $a^n$ - $b^n$ is divisible by (a+b) and (a-b) when n is an even number.

$897^{43}$ - $103^{43}$ is divisible by 897 + 103 = 1000 and 897 - 103 = 794.

Now, 1000 = $2^3 \times 5^3$ and 794 = 2 × 397

We see that $5^3$ is a factor of 1000, therefore the number must also be divisible by 125

**Video Solution:**

**Q20 Text Solution:**

$2370^{5320}$ will have 5320 0's at its end. Hence the rightmost nonzero digit will be same as the unit digit of $237^{5320}$.

Since 5320 is divisible by 4, so the unit digit will be from the fourth cycle of 7's cyclicity.

So the rightmost non-zero digit will be 1.

**Video Solution:**

**Q21 Text Solution:**

$354x\times 7541\times 4355\times 8789\times 5653$
$=354x\times ****5$

So the unit digit will be the unit digit of $x\times 5$.

Since the unit digit is 5, this means $(x\times 5)$ has 5 as unit digit, which is possible when x is odd, so there are 5 posssible values of x, i.e., 1, 3, 5, 7 and 9.

**Video Solution:**

**Q22 Text Solution:**

Considering only the unit digit's

$$38^{47}-26^{39}-17^{13}$$
$$=8^{47}-6^{39}-7^{13}$$
$$=8^{11(4)+3}-6^{39}-7^{3(4)+1}$$

Cyclicity of 8 = 8, 4, 2, 6

Cyclicity of 6 = 6

Cyclicity of 7 = 7, 9, 3, 1

So the numbers are going to be of the form

(****2)-(****6)-(****7)

=(****9)

So the unit digit will be 9.

**Video Solution:**

**Q23 Text Solution:**

Unit digit is going to be same for

$41^2,\ 51^2,\ 61^2,\ 71^2,\ 81^2$ *and* $91^2$, which will be same as that of $1^2$.

Similarly for other numbers it is also going to be same, i.e., $2^2,\ 3^2,\ 4^2$ and so on uptil $0^2$.

So

$$6\times\left(unit\ digit\ of\ 1^2+2^2+3^2+\ldots+9^2+0^2\right)$$
$$=6\times\left(unit\ digit\ of\ 1+4+9+6+5+6+9+4+1+0\right)$$
$$=6\times\left(unit\ digit\ of\ 45\right)$$
$$=6\times5$$
$$=30$$

So unit digit is going to be 0.

**Video Solution:**

**Q24 Text Solution:**

Both 3 and 7 have a cycle of 4.

Difference between (n-1) and (n+3) is 4, so if both of them are divided by 4, they will provide us with same remainder.

So if (n-1) and (n+3) are of format 4x, then

$$Unit\ digit\ of\ \left[3^{(n-1)}\times7^{(n+3)}\right]$$
$$=unit\ digit\ of\ 3^4\times7^4$$
$$=unit\ digit\ of\ 1\times1$$
$$=1$$

If (n-1) and (n+3) are of format 4x+1, then

$$Unit\ digit\ of\ \left[3^{(n-1)}\times7^{(n+3)}\right]$$
$$=\ unit\ digit\ of\ 3^1\times7^1$$
$$=unit\ digit\ of\ 3\times7$$
$$=1$$

If (n-1) and (n+3) are of format 4x+2, then

$$Unit\ digit\ of\ \left[3^{(n-1)}\times7^{(n+3)}\right]$$
$$=\ unit\ digit\ of\ 3^2\times7^2$$
$$=unit\ digit\ of\ 9\times9$$
$$=1$$

If (n-1) and (n+3) are of format 4x+3, then

$$Unit\ digit\ of\ \left[3^{(n-1)}\times7^{(n+3)}\right]$$
$$=\ unit\ digit\ of\ 3^3\times7^3$$
$$=unit\ digit\ of\ 7\times3$$
$$=1$$

Hence irrrespective of the value of n, the unit digit is always going to be 1.

**Video Solution:**

**Q25 Text Solution:**

Multiplying the unit digits, it will come out to be

$3 \times 9 \times 7 \times 1 \times 3 \times 9 \times \ldots$

The cycle of 4 multiplicands will be repeated. So for 111 terms, 4 multiplicands will be repeated 27 times and then next three terms will be multiplied, so

$(3 \times 9 \times 7 \times 1)^{27} \times 3 \times 9 \times 7$

$= (**9)^{27} \times \left(**9\right)$

So considering only the unit digits, it will become

$9^{27} \times 9$

$= 9^{28}$

Even power of 9 gives us unit digit as 1.

**Video Solution:**

**Q26 Text Solution:**

$625^{33.75} \times 81^{12.25} \times 49^{11.5}$

$= 625^{33} \times 625^{\frac{3}{4}} \times 81^{12} \times 81^{\frac{1}{4}} \times 49^{11} \times 49^{\frac{1}{2}}$

$= 625^{33} \times 125 \times 81^{12} \times 3 \times 49^{11} \times 7$

The unit digit here will be same as unit digit of

$5^{33} \times 5 \times 1^{12} \times 3 \times 9^{11} \times 7$

$= 5^{34} \times 3 \times 9^{11} \times 7$

So the unit digits will be 5, 3, 9 and 7 respectively, so

$5 \times 3 \times 9 \times 7$

$= ***5$

**Video Solution:**

**Q27 Text Solution:**

Unit digit of $27^{21!^{31!}}$ will be same as $7^{21!^{31!}}$.

Cyclicity of 7 = 7, 9, 3, 1

Now we need to check the power of 7 with respect to 4.

However 21! is divisible by 4 (All factorials >3 are divisible by 4), and hence all its powers will be divisible by 4, so

$7^{21!^{31!}} = 7^{4x}$

Hence the unit digit will be 1.

**Video Solution:**

**Q28 Text Solution:**

$\frac{24^{42}}{6^{24}} + \frac{16^{60}}{64^{32}}$

$= \frac{6^{42} \times 4^{42}}{6^{24}} + \frac{2^{240}}{2^{192}}$

$= 6^{18} \times 4^{42} + 2^{48}$

Unit digit for any power of 6 is always 6 and unit digit for even power of 4 is 6.

Cycle of 2 = 2, 4, 8, 6

So

$6^{18} \times 4^{42} + 2^{48}$

$= ***6 \times ***6 + ***6$

$= ***6 + ***6$

$= ***2$

So the unt digit will be 2.

**Video Solution:**

**Q29 Text Solution:**

$$a=\frac{27^{52}}{3^{5}}+\frac{16^{36}}{4^{6}}$$
$$=\frac{(3)^{3\times52}}{3^{5}}+\frac{16^{36}}{16^{3}}=3^{(156-5)}+16^{(36-3)}$$
$$=3^{151}+16^{33}$$

Unit digit of $3^{151}=7 \quad \left(3^{4m+3}\text{ type }\right)$

Unit digit of $16^{33}=6 \quad \left(6^{\text{Any power}}=6\right)$

So, $6+7\rightarrow3$ is the last digit of the given expression.

**Video Solution:**

**Q30 Text Solution:**

6 raised to any power ends in 6

4 raised to odd power ends in 4

5 raised to any power ends in 5

6 + 4 + 5

= 15

Thus, the units digit will be **5**

**Video Solution:**